POWER BI DATA ANALYTICS PORTFOLIO

Olympic Insights using Data Analytics

Milestone Project 3 | Created by Mohd Isaar |
Data source: Olympic Games dataset (WsCube Tech)

DATASET SCOPE

1896–2022

Historical modern Olympiads

TOTAL MEDALS ANALYSED

20,412

Cleaned medal outcomes

PRIMARY ANALYTICS TOOL

Power BI

DAX · Power Query modeling

PORTFOLIO PROJECT · POWER BI & DAX · COMPREHENSIVE OLYMPIC ANALYSIS

PROJECT FOUNDATIONS

Objective, Data and Tools

GOALS & STAKEHOLDERS

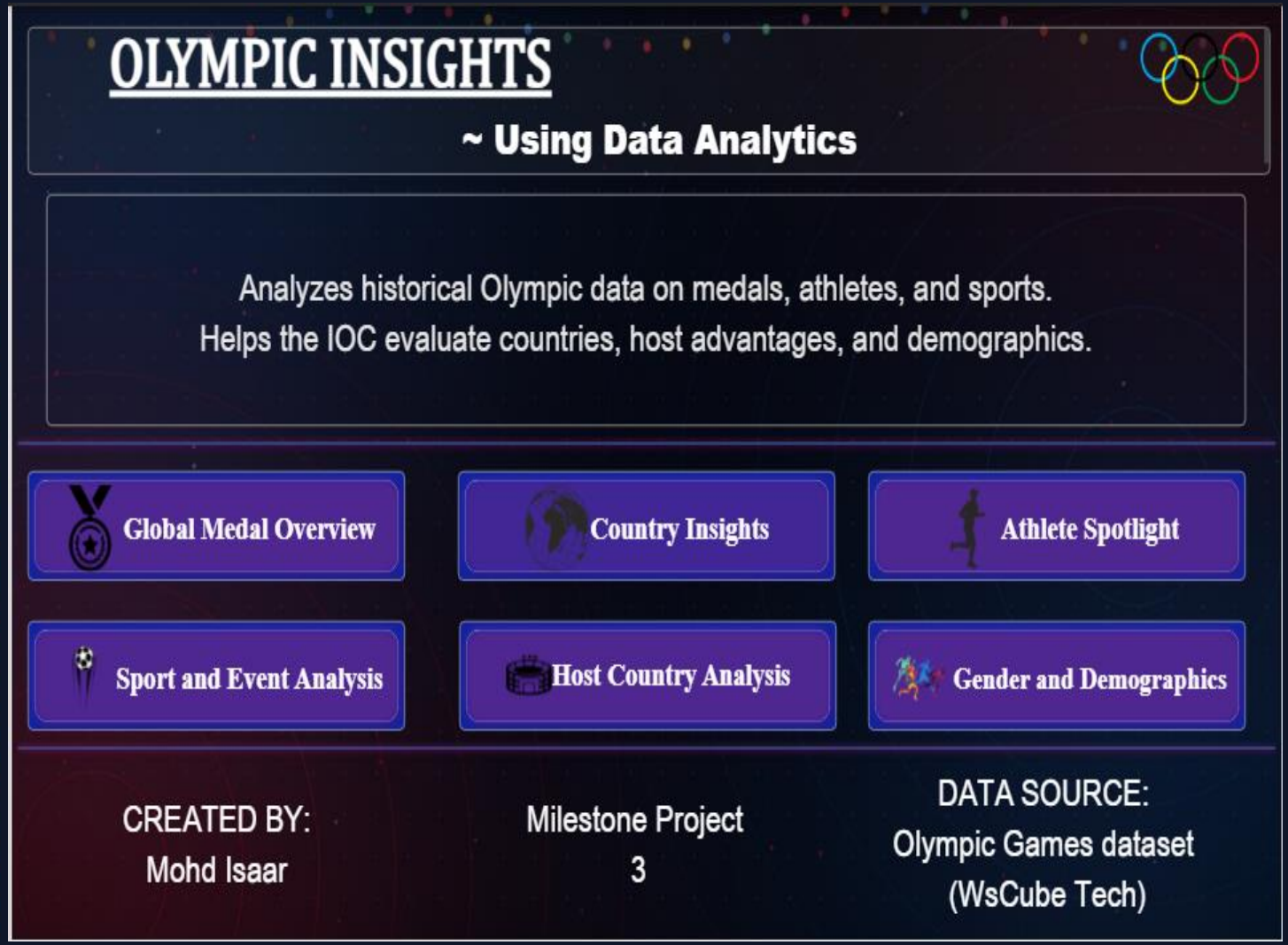
- Analyse Olympic data from 1896 to 2022: medals, athletes, sports and host countries.
- Help the IOC (International Olympic Committee) see which countries perform best and how hosting and demographics matter.

CORE QUESTIONS & METHODOLOGY

- Questions: Who wins the most medals? Does hosting help? How do gender and age change over time?
- Tools: Power BI, DAX and Power Query.

Home Page

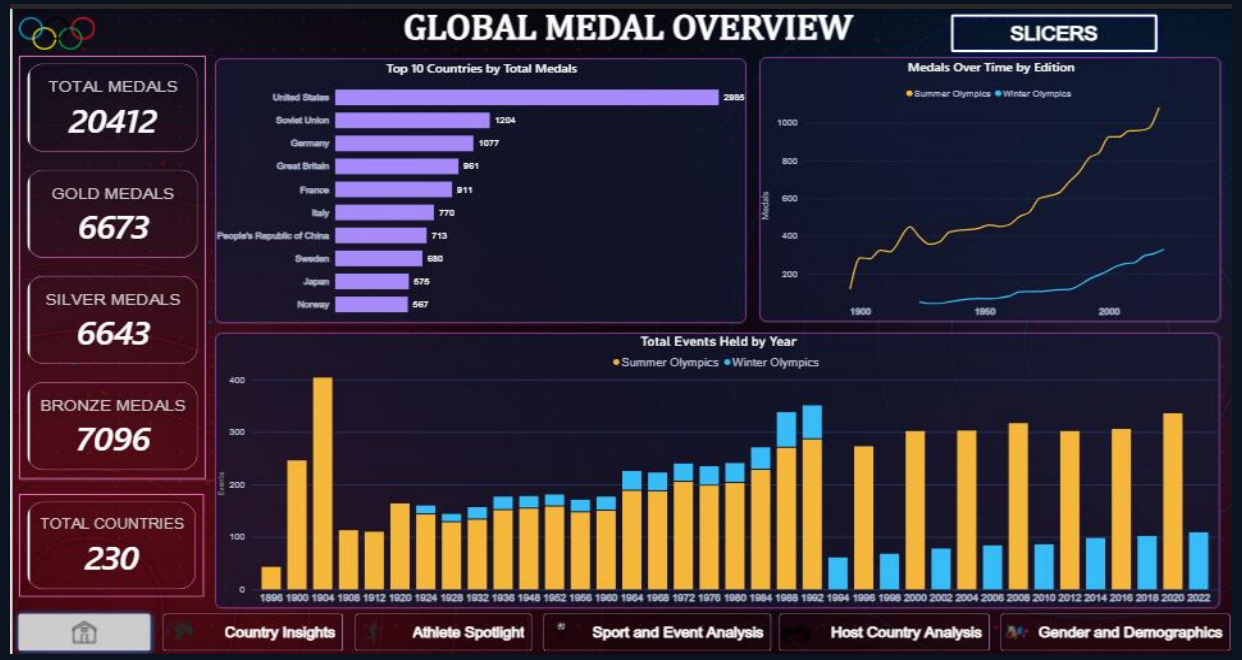
- Interactive Power BI dashboard with 7 pages: Home plus 6 analysis pages.
- Navigation buttons on every page.
- Slicers: Season, Year, Sport, Country and Host Country.
- Pages: Global Medal Overview, Country Insights, Athlete Spotlight, Sport and Event Analysis, Host Country Analysis, Gender and Demographics.

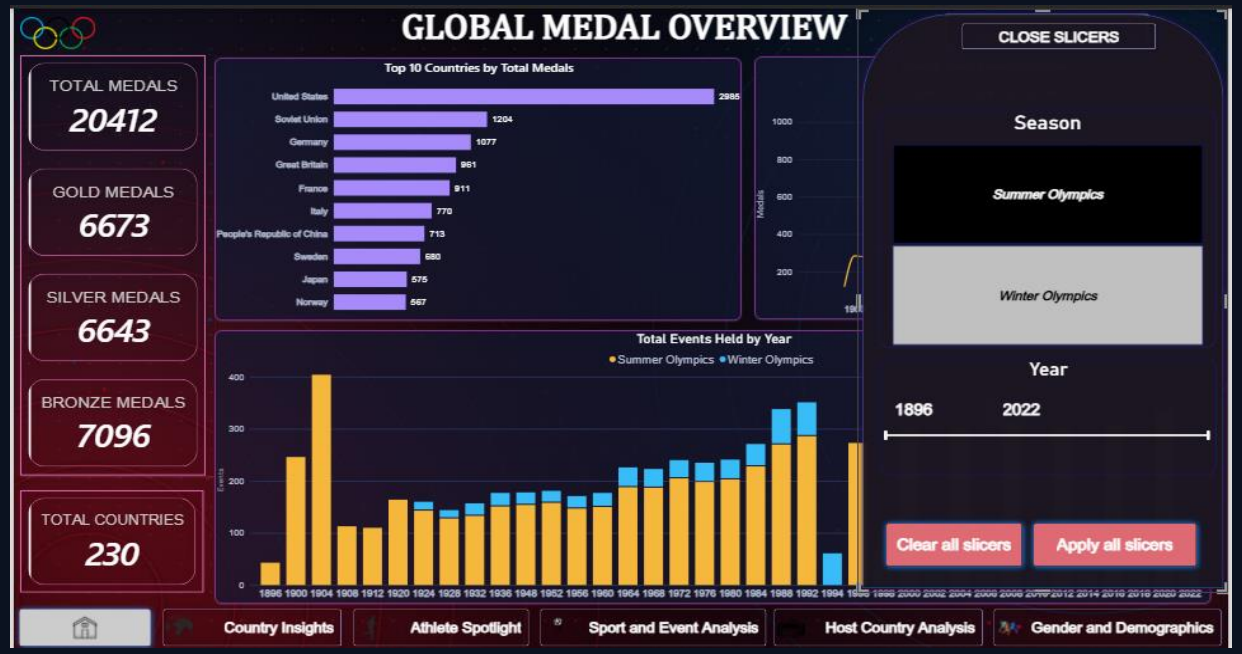


REPORT PAGE 01

Global Medal Overview

- 20,412 medals: 6,673 gold, 6,643 silver, 7,096 bronze.
- 230 participating countries.
- Top countries: United States 2,985, Soviet Union 1,204, Germany 1,077.
- Summer medals have grown much faster than Winter medals.
- Insight: the three medal types are almost equal, with bronze slightly higher.

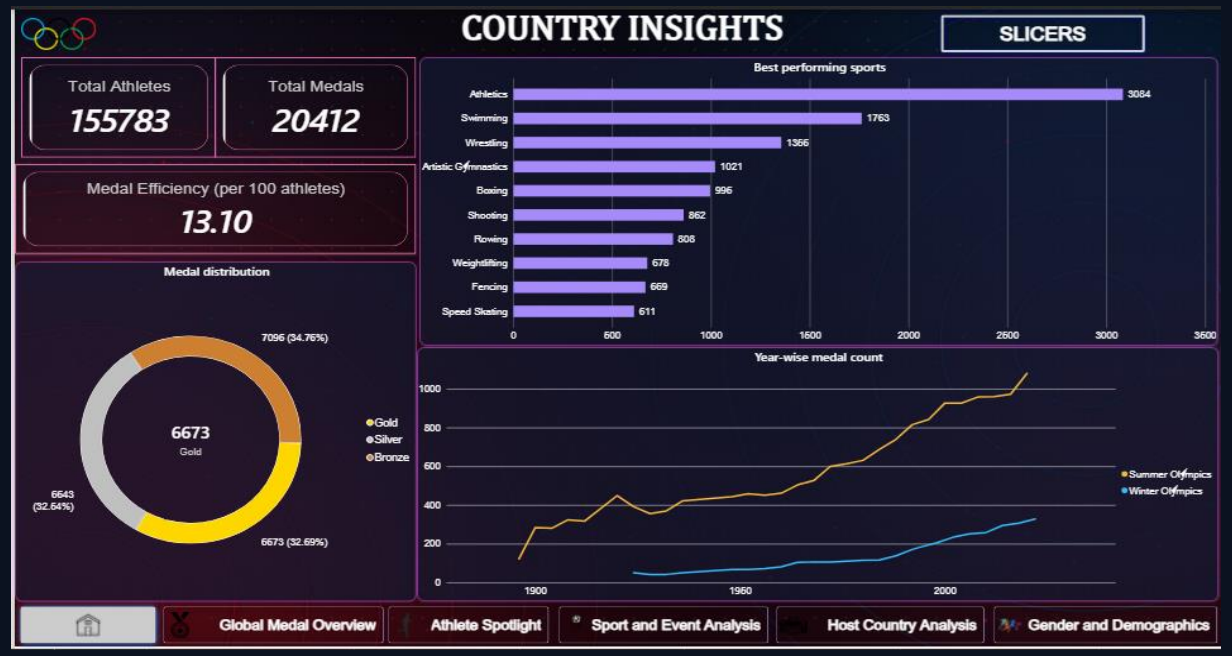


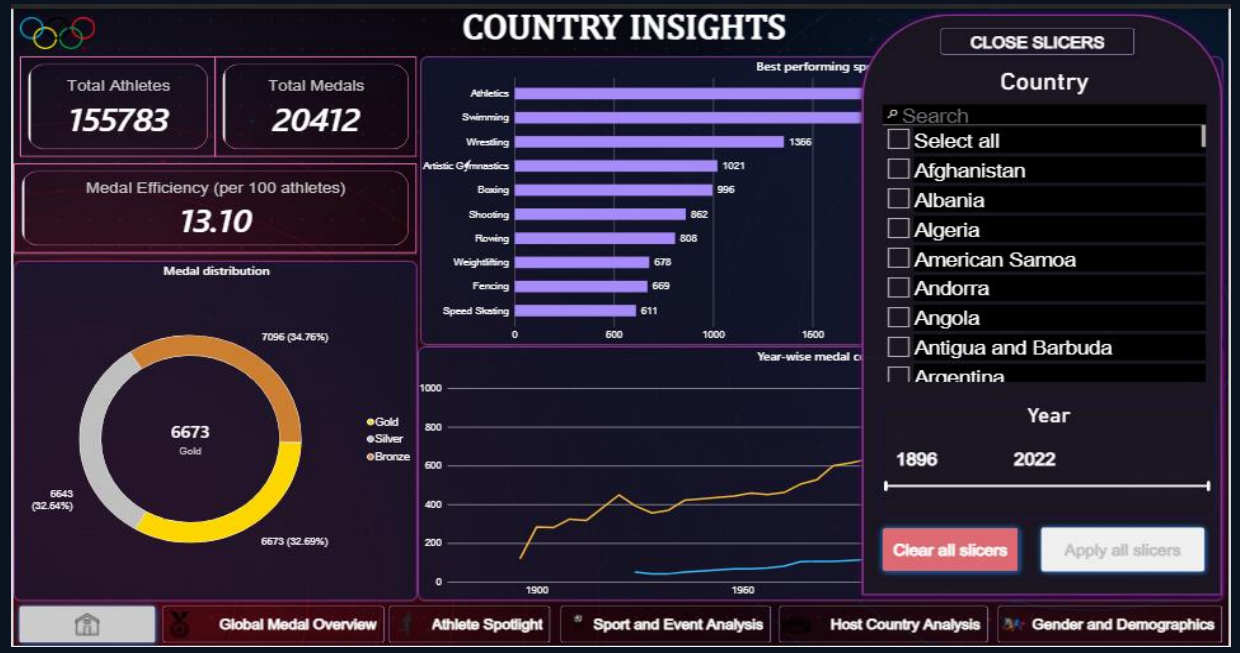


REPORT PAGE 02

Country Insights

- 155,783 athletes won 20,412 medals.
- Medal efficiency: 13.10 medals per 100 athletes.
- Top sports: Athletics 3,084, Swimming 1,763, Wrestling 1,356.
- Medal split: gold 32.7%, silver 32.5%, bronze 34.8%.
- Use the Country slicer to see any one country.

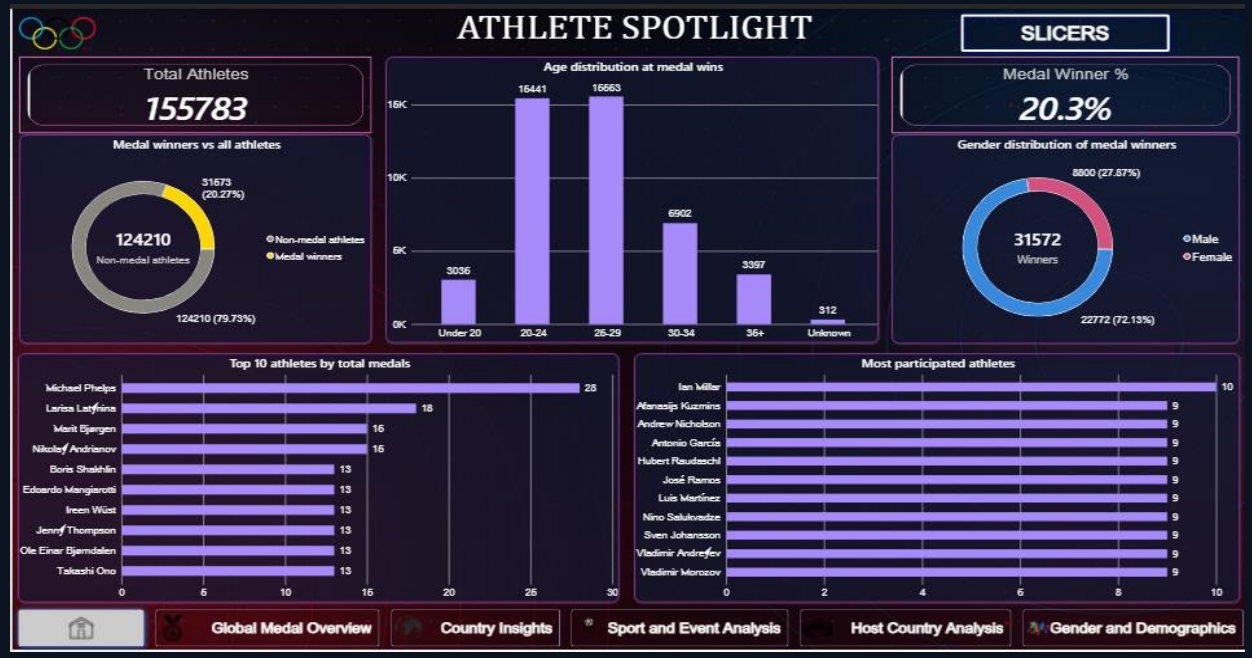


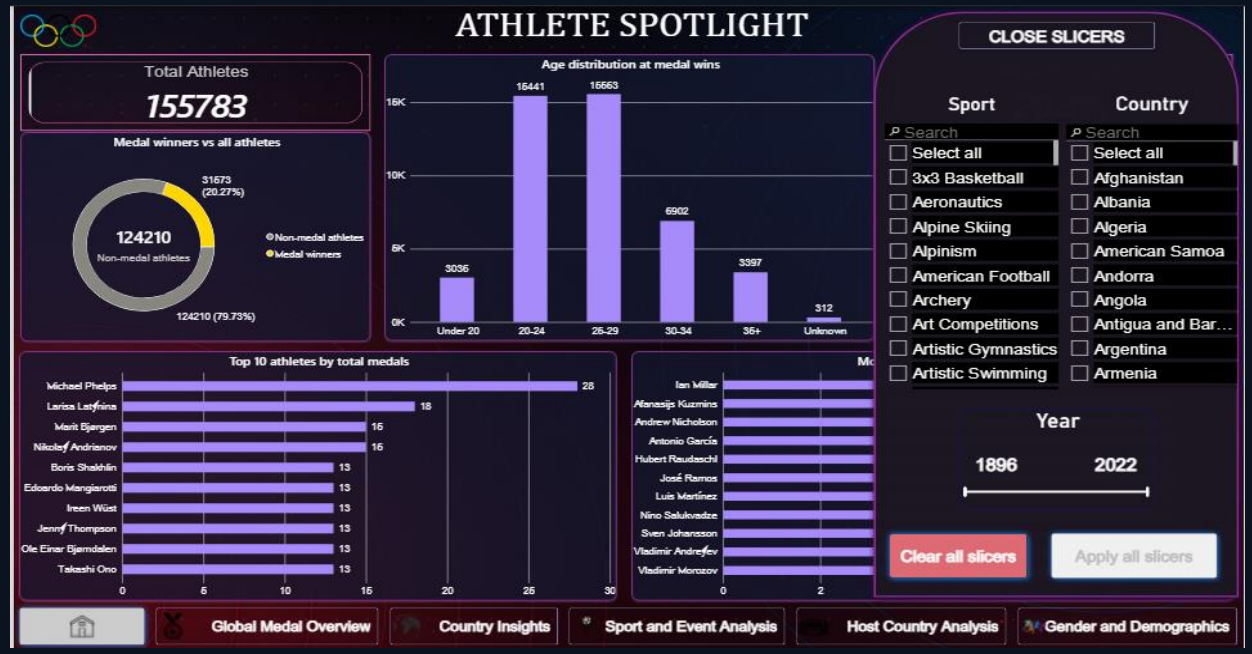


REPORT PAGE 03

Athlete Spotlight

- Michael Phelps leads with 28 medals, then Larisa Latynina with 18.
- About 20.3% of athletes (31,573) won at least one medal.
- Most medal wins come at ages 20-29.
- Medal winners are about 72% male and 28% female.
- Note: medal winners are more than total medals because a team medal counts for every team member.

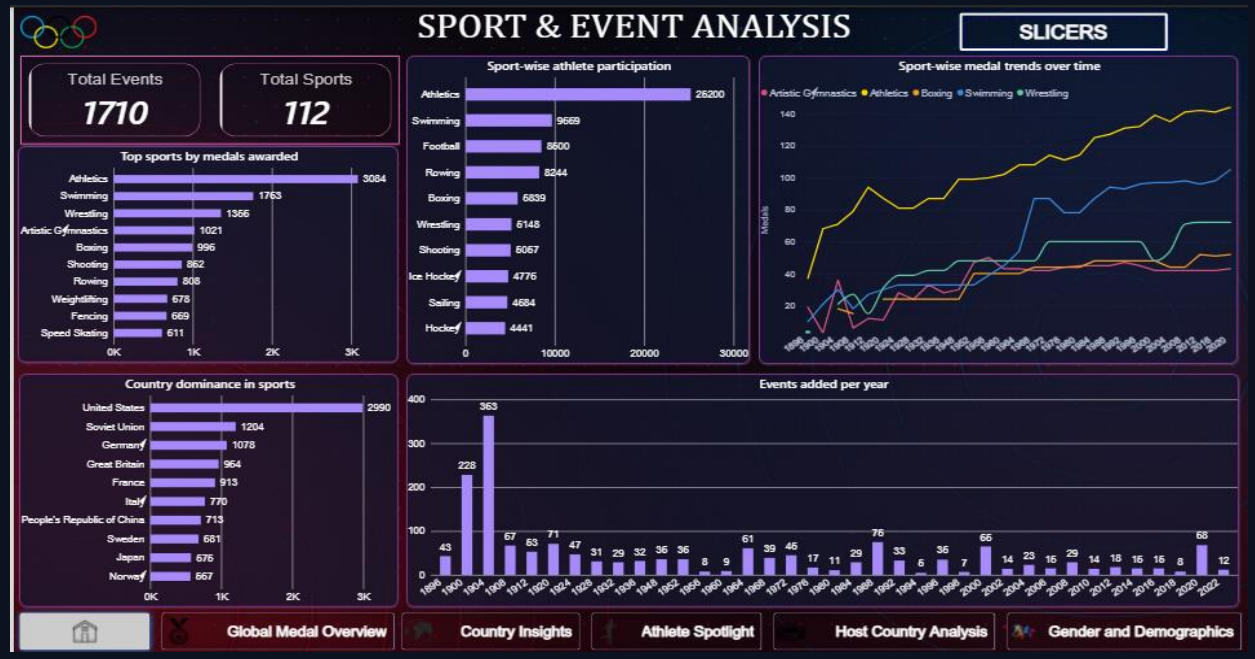


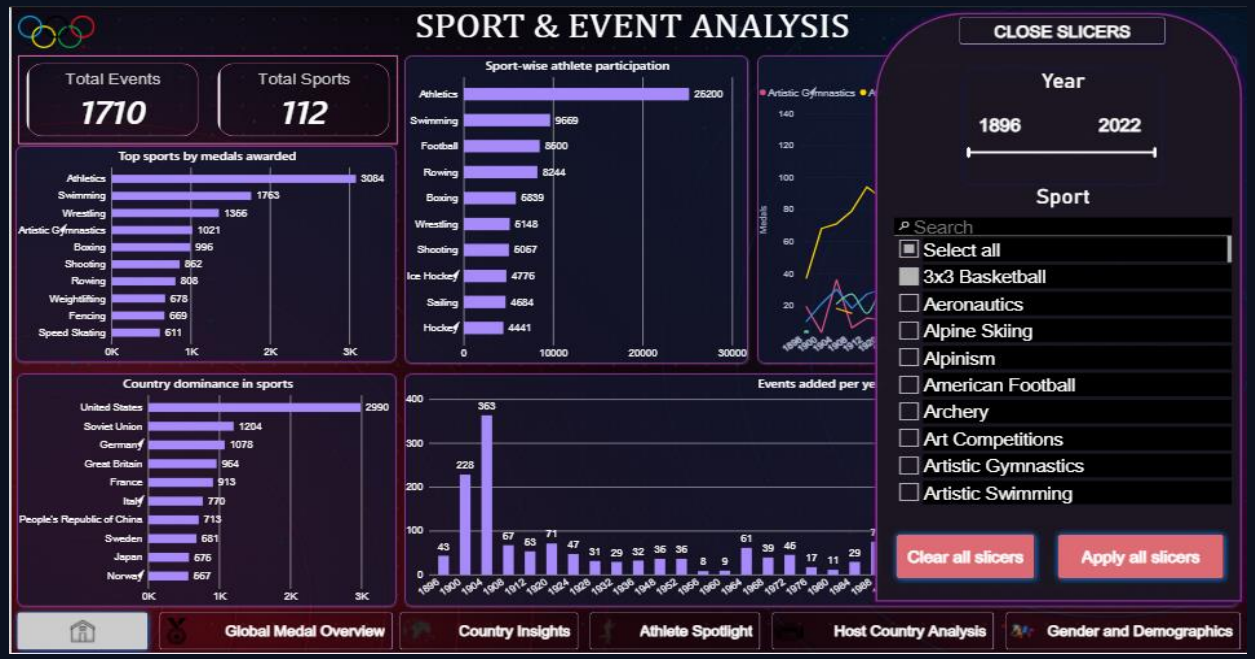


REPORT PAGE 04

Sport and Event Analysis

- 1,710 events and 112 sports.
- Athletics has the highest participation (about 26,200), then Swimming and Football.
- Events added per year peaked in 1904 (363 events).
- Note: the 1900 and 1904 counts look high because events are counted by name, so variants of the same event count separately.

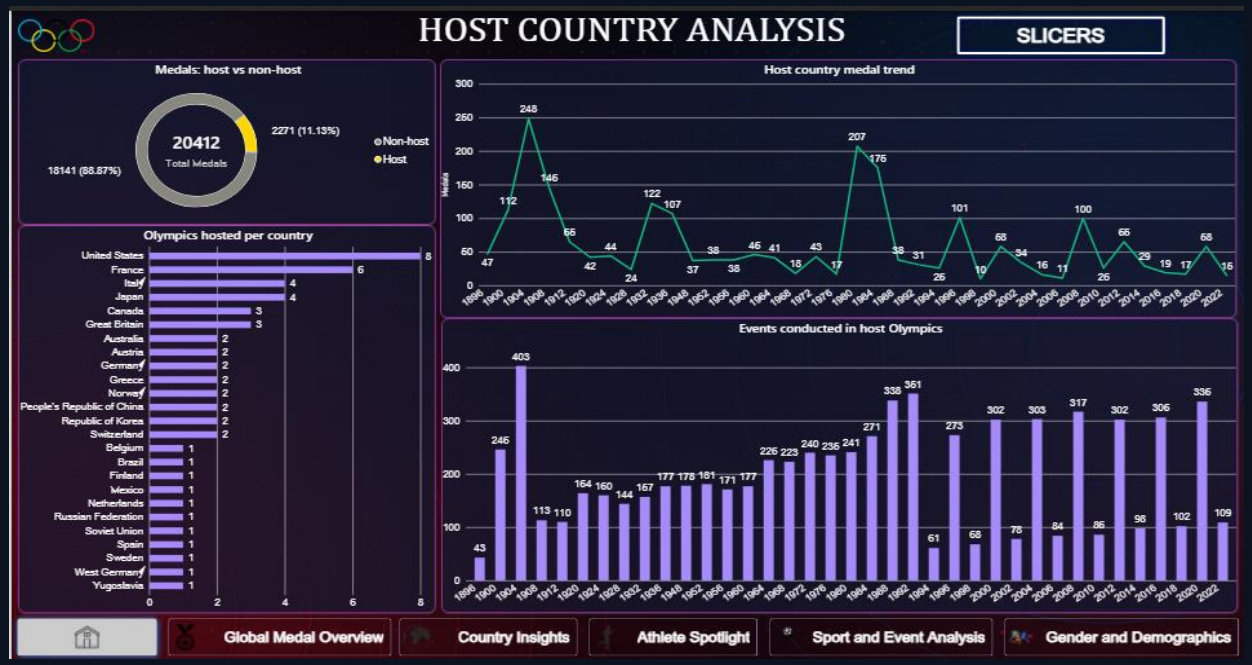


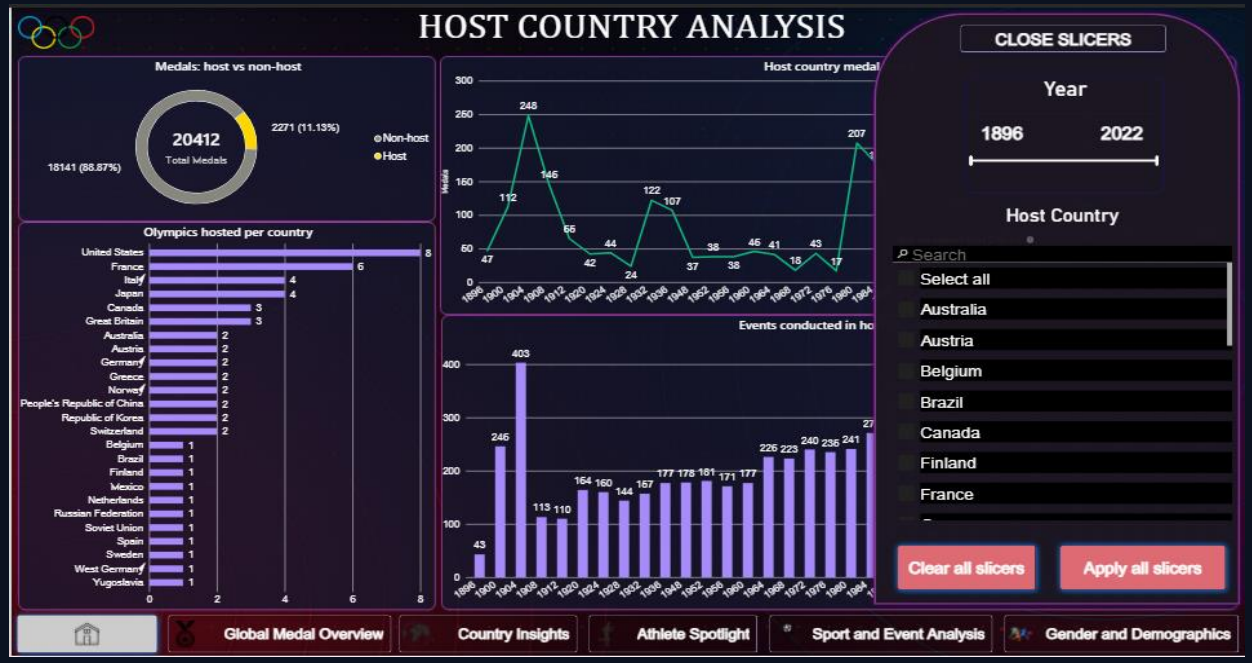


REPORT PAGE 05

Host Country Analysis

- Host countries won 11.13% of medals (2,271 of 20,412).
- Non-host countries won 88.87% (18,141).
- United States has hosted the most times (8), then France (6).
- Host-year medal counts peaked in 1984 and 1904.
- Note: Germany and West Germany are counted as separate hosts.

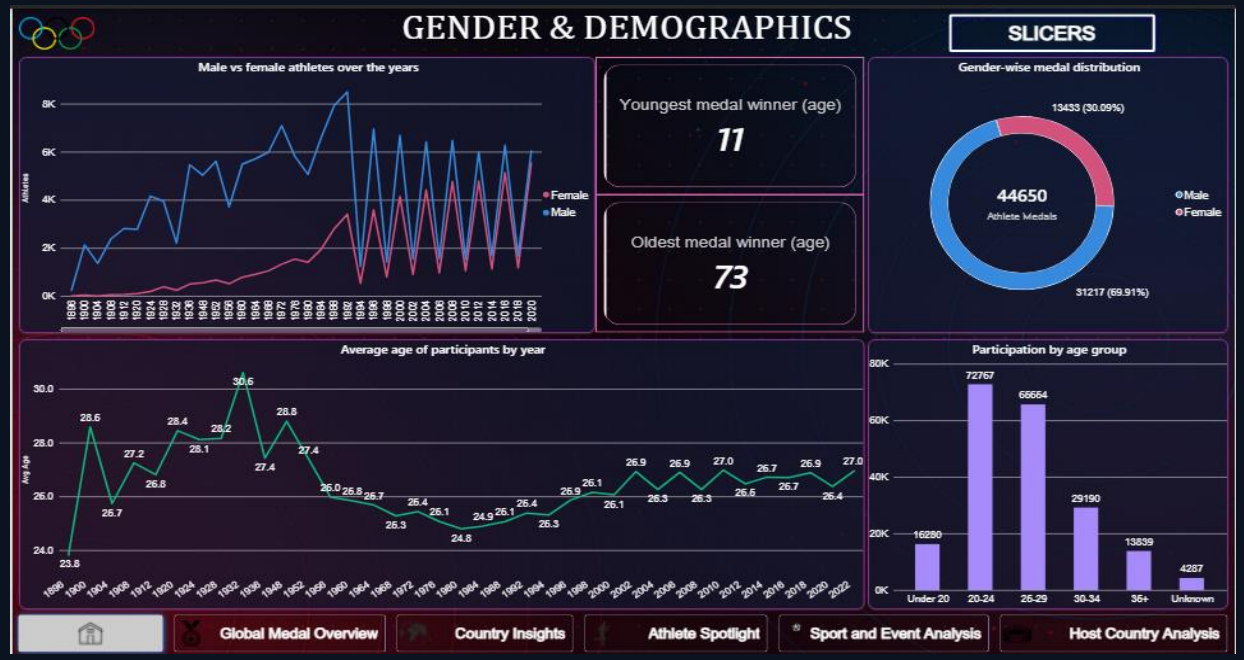


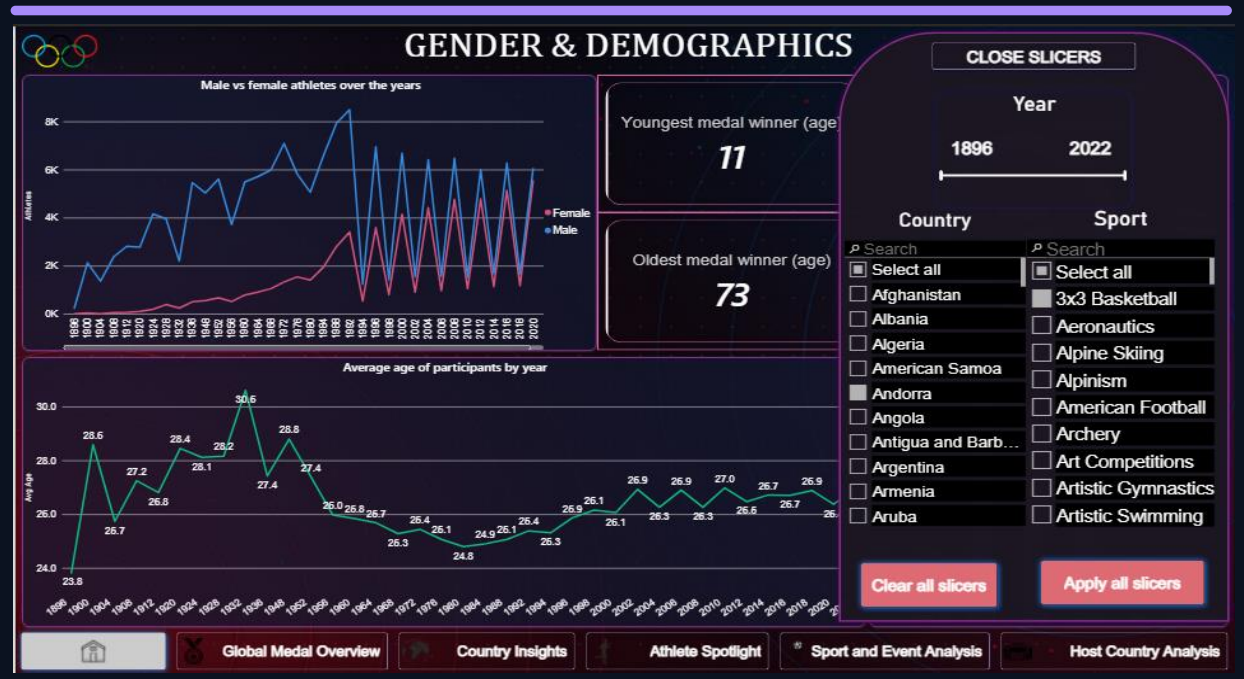


REPORT PAGE 06

Gender and Demographics

- About 30% of athlete medals went to women (13,433 of 44,650).
- Male participation is higher but the gap is narrowing.
- Youngest medal winner is 11, oldest is 73.
- Average athlete age is about 26-27 in recent Games.
- Note: 44,650 is more than 20,412 because a team medal counts for every team member.
- Note: the zigzag in the gender line chart is real. Summer and Winter Games fall in different years after 1992.





SYNTHESIS

Key Takeaways

WHAT THE DATA SHOWS

- Medals are concentrated in a few countries: United States 2,985, Soviet Union 1,204, Germany 1,077.
- Athletics (3,084) and Swimming (1,763) award the most medals.
- Summer medals have grown much faster than Winter medals.
- Gold, silver and bronze are almost equal (32.7%, 32.5%, 34.8%).

PEOPLE AND PLACES

- Hosting gives a small medal share (11.13%), not a dominant one.
- Female participation is increasing, with about 30% of athlete medals going to women.
- Most medal wins come at ages 20-29.
- About 20.3% of athletes won at least one medal.

FUTURE SCOPE

Challenges and Next Steps

CHALLENGES SOLVED

- Removed the 1906 Intercalated Games and the 1956 Equestrian events so totals match on every page (20,412 medals).
- Handled team medals: the tally counts a team medal once, but athlete-level pages count every team member.
- Connected the country slicer by activating the relationship between the tables.

NEXT STEPS

- Add predictive analysis of future medal counts.
- Add more recent data (Paris 2024 and Milano Cortina 2026).
- Merge historic countries (USSR, West Germany) with their successors for fairer country comparisons.

GitHub link: https://github.com/isaar1/Milestone_Project-Olympic_Insights